Restaurant Sales & Operations Analytics Project Documentation

PART 1

Project Introduction

1.1 **Project Overview**

The Restaurant Sales & Operations Analytics Project was carried out to analyze the sales performance, customer behavior, operational efficiency, and overall business performance of a fast-growing restaurant chain. The project involved working with a messy real-world dataset containing restaurant transactions, customer records, sales details, and operational information. The analysis process involved using Microsoft Excel, Power Query, SQL, and Power BI to transform raw business data into meaningful insights that can support business decision-making. The project focused on exploratory data analysis, business intelligence reporting, dashboard creation, and insight generation. This project simulates the responsibilities of a real-world Data Analyst by requiring the identification of trends, patterns, business opportunities, and operational challenges within the restaurant business.

1.2 Project Objectives

- The major objectives of this project include:
- To analyze restaurant sales and operational performance.
- To identify sales trends and customer purchasing behavior.
- To evaluate restaurant and product performance.
- To generate insights that support strategic business decisions.
- To build interactive dashboards for data visualization.
- To improve analytical and business intelligence skills using Excel, SQL, Power BI, and Power Query.
- To create a portfolio-ready analytics project.

1.3Business Problem

The restaurant management team faced several business challenges due to the lack of organized and actionable insights from their operational data. Some of the major business problems included:

- Difficulty identifying top-performing products and branches.
- Inability to track customer spending patterns effectively.
- Lack of visibility into revenue trends over time.
- Presence of inconsistent and messy records affecting reporting accuracy.
- Difficulty detecting suspicious or duplicate transactions.
- Limited insight into operational inefficiencies.

This project was designed to solve these problems by transforming the raw data into clear visual reports and actionable business insights.

PART 2

Dataset Description

2.1 **Dataset Name**

Capstone messy_restaurant_data

2.2 Dataset Overview

The dataset contains restaurant transaction and operational records collected from different branches of the restaurant business. The dataset includes customer, restaurant, dish ordered, rating, date of visit, feedback, price of dish ordered, quantity.

2.3 Number of Rows and Columns

The dataset contains forty-five rows representing restaurant transactions and seven columns representing different business variables such as:

- Customer
- Restaurant
- Dish_ordered
- Rating
- Date_of_Visit
- Feedback
- Price_of_dish_ordered
- Quantity

2.4 Column Descriptions

COLUMN NAME	**DESCRIPTION**
Customer Name	Name of the customer who placed the order
Restaurant	Restaurant where transaction occurred
Dish ordered	Name of the food product sold
Rating	Representation of customer’s review of dish ordered on a positive number scale
Date of visit	Date when the transaction took place
Feedback	Representation of customer’s review of dish ordered using designated phases
Price of dish ordered	Cost of dish ordered per quantity ordered
Quantity	Number of items purchased

2.5 Data Quality Issues Discovered

During the inspection of the dataset, several data quality issues were discovered, including:

Missing values in important fields.

Empty rows and blank cells.

Leading and training spaces.

Outliers in sales amounts and quantities.

These issues affected data consistency and reliability before analysis.

PART 3

Data Cleaning

3.1 Cleaning Steps Performed

- The cleaning steps performed include:
- Checking for duplicates.
- Trimming leading and training spaces.
- Filling up empty cells.

3.2 Problems Encountered

- The problems encountered include:
- Leading and Training Spaces
- Empty Cells/ Missing values

3.3 Solutions Implemented

Trimming Extra Spaces: the TRIM function was used to removing trimming, leading and unnecessary spaces from text fields especially in customer name, restaurant and dish_ordered.

Filling Empty Spaces:

- Missing or blank spaces were handled by:
- Replacing blanks with appropriate values where necessary.
- Using suitable default values.
- Ensuring important cells did not remain empty.

PART 4

Analysis Process

4.1 SQL Analysis Approach

SQL was used to query and analyze the restaurant dataset efficiently. SQL queries were written to answer major business questions related to sales performance, customer behavior, branch performance, and product analysis. The SQL analysis process involved:

Importing the cleaned dataset into a relational database.

Creating database tables and defining appropriate data types.

Writing SELECT statements for data exploration.

Using aggregate functions such as: SUM(), AVG(), COUNT(), MAX(), MIN()

Applying GROUP BY and ORDER BY clauses for categorization and ranking.

Identifying trends and operational insights.

Examples of business questions answered using SQL include:

- Total revenue generated.
- Top-performing branches.
- Highest-selling products.
- Customers with the highest spending.
- Monthly and daily sales trends.
- Revenue by restaurant.
- Weekday versus weekend sales comparison.

SQL made it possible to perform fast and accurate business analysis on large volumes of restaurant transaction data.

4.2 Excel Analysis Approach

Microsoft Excel was used for exploratory data analysis and business reporting. Excel tools helped in summarizing large amounts of data into understandable business insights. The Excel analysis process involved:

- Creating Pivot Tables for summarization.
- Using Pivot Charts for visual analysis.
- Applying formulas and functions such as: SUM, AVERAGE, AVERAGEIF
- Creating KPIs for: Total Revenue, Total Quantity
- Building slicers for interactive filtering.
- Performing trend analysis using charts.
- Comparing restaurant and product performance.

Excel provided a user-friendly environment for understanding business trends and generating quick insights.

3.3 Power BI Analysis Approach

Power BI was used to create a professional and interactive business intelligence dashboard. The dashboard helped visualize key metrics and operational insights.

The Power BI process involved:

- Importing the cleaned dataset into Power BI.

- Creating relationships between tables where necessary.
- Creating DAX measures for KPI calculations.
- Designing interactive dashboard pages.
- Using visuals such as:
- KPI Cards
- Bar Charts
- Line CharDonut ChartsTreemapsMatrix TablesSlicers
- Performing time-series analysis.
- Creating drill-down reports for detailed exploration.

4.4 DAX Measures Created

Some important DAX measures created include:

- Total Revenue
- Average Revenue
- Total Orders
- Total Customers
- Monthly Revenue
- Revenue Growth
- Product Performance Metrics

Power BI improved data storytelling by presenting business performance in a visually appealing and interactive format.

4.5 Key Insights

Important Discoveries

The analysis revealed several important findings about the restaurant business:

- Certain food items generated significantly higher revenue compared to others.
- Some products recorded consistently low sales performance.
- A few branches contributed the majority of total revenue.
- Some customers repeatedly generated high transaction values.
- Sales performance varied across different months and weekdays.
- Certain transaction records appeared duplicated or suspicious.
- Revenue trends showed seasonal and time-based fluctuations.

Trends Identified

Several business trends were identified during the analysis:

Sales Trends

- Revenue increased during peak business periods.
- Weekend sales were generally higher than weekday sales.
- January generated significantly higher revenue than February.

Customer Trends

- Repeat customers contributed a large portion of revenue.
- High-value customers had larger average order sizes.
- Certain cities generated more customer activity than others.

Product Trends

- Top-selling products consistently contributed most of the revenue.
- Some product categories performed better than others.
- Low-performing products reduced overall profitability.

Operational Trends

- Certain branches consistently underperformed.
- Duplicate and unusual transactions indicated possible operational issues.
- Inconsistent records affected reporting accuracy.

Recommendations

Based on the analysis, the following recommendations were made:

Product Recommendations

- Promote top-selling and high-profit products.
- Introduce marketing campaigns for high-performing categories.
- Review or discontinue consistently low-performing products.

Customer Recommendations

- Implement customer loyalty programs for repeat buyers.

- Target high-value customers with special offers and discounts.
- Improve customer engagement strategies.

Restaurant Recommendations

- Investigate operational challenges in underperforming restaurant.
- Replicate strategies used by high-performing locations.
- Improve inventory and staffing management.
- Each restaurant should pay detailed-attention to their product that generates the most revenue

Operational Recommendations

- Improve transaction monitoring systems.
- Standardize data entry processes.
- Reduce suspicious transactions.
- Improve data management and reporting processes.

Business Intelligence Recommendations

- Continue using Power BI dashboards for monitoring performance.
- Implement real-time reporting systems.
- Use data-driven decision-making for future business planning.

PART 5

Conclusion

Summary of Finding

This project successfully transformed raw restaurant operational data into meaningful business insights using Excel, SQL, Power Query, and Power BI. The analysis helped identify top-performing products, valuable customers, sales trends, branch performance, and operational issues.

The project also demonstrated how business intelligence tools can improve decision-making and operational efficiency in the restaurant industry.

Interactive dashboards and visual reports made it easier to understand complex data patterns and communicate findings effectively.

Final Business Recommendations

The restaurant management team should:

Focus more on high-performing products and restaurants.

Improve customer retention strategies.

Monitor operational inefficiencies closely.

Invest in data-driven decision-making processes.

Use interactive dashboards for regular performance monitoring.

Improve data quality management practices.

Continuously analyze sales and customer trends to maximize profitability.

Overall, the project showed the importance of data analytics in helping businesses improve profitability, customer satisfaction, and operational performance.